WORKED SOLUTION

STEP I 1991, Question 6

Pure • Stationary Points and Improper Integrals

STEP I 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q6

This question has a full guided treatment in *Beyond STEP*, Section 3.1, *Spot the Flaw*: https://beyondstep.keithmcnulty.org/ch03/spot-the-flaw.html

The key idea

Don't trust anything: redo each calculation yourself and ask, at each "hence" or "thus", whether the step really follows. In (i) the derivative has been computed wrongly, and even with the right stationary points the reasoning about maxima, minima and global bounds would be faulty. In (ii) the antiderivative is wrong, and, more seriously, the integrand blows up inside the interval, so the fundamental theorem of calculus cannot be applied at all.

Solution

Part (i)

Step 1: the stationary points. By the quotient rule,

$$g'(x) = \frac{6x^2(x^4+4) - 4x^3(2x^3+3)}{(x^4+4)^2} = \frac{-2x^6 - 12x^3 + 24x^2}{(x^4+4)^2} = \frac{-2x^2\,(x^4+6x-12)}{(x^4+4)^2}.$$

So $g'(x) = 0$ exactly when $x = 0$ or $h(x) = 0$, where $h(x) = x^4 + 6x - 12$. Now $h(1) = -5$ and $h(-1) = -17$, so $x = \pm 1$ are *not* stationary points: in fact $g'(1) = \frac{2}{5}$ and $g'(-1) = \frac{34}{25}$. The claim fails at the first step.

How many roots does h have? Since $h'(x) = 4x^3 + 6$ vanishes only at $x_0 = -(3/2)^{1/3}$, h decreases and then increases, so it can cross zero at most twice. Also $h(0) = -12 < 0$ and $h(x) \to +\infty$ as $x \to \pm\infty$, so it crosses at least twice. Hence h has exactly two real roots, $\beta < 0 < \alpha$. Since $h(1) < 0 < h(2) = 16$ and $h(-3) = 51 > 0 > h(-2) = -8$,

$$1 < \alpha < 2, \qquad -3 < \beta < -2,$$

and numerically $\alpha \approx 1.386$, $\beta \approx -2.247$.

Step 2: the stationary values. $g(0) = \frac{3}{4}$ is correct, and $g(1) = 1$ is correct too, but $g(-1) = \frac{1}{5}$, not 1 (and neither point at ± 1 is stationary anyway). At a root t of h we have $t^4 = 12 - 6t$, so $t^3 = (12 - 6t)/t$ and

$$g(t) = \frac{2(12-6t)/t + 3}{16 - 6t} = \frac{24 - 9t}{t(16-6t)} = \frac{3(8-3t)}{2t(8-3t)} = \frac{3}{2t},$$

cancelling $8 - 3t \neq 0$ (as $t = \frac{8}{3}$ is not a root of h). So the stationary values are $g(0) = \frac{3}{4}$, $g(\alpha) = \frac{3}{2\alpha} \approx 1.083$ and $g(\beta) = \frac{3}{2\beta} \approx -0.668$.

Ask yourself If one stationary value is smaller than another, must the smaller one be a minimum?

Step 3: the nature of the stationary points. No. Comparing the *values* at stationary points tells us nothing about their nature; we must look at the sign of g' on either side (or at g''). From the factorised form, the sign of $g'(x)$ is the sign of $-h(x)$ for $x \neq 0$, since $x^2 > 0$ and $(x^4 + 4)^2 > 0$. As $h < 0$ on (β, α) and $h > 0$ outside $[\beta, \alpha]$:

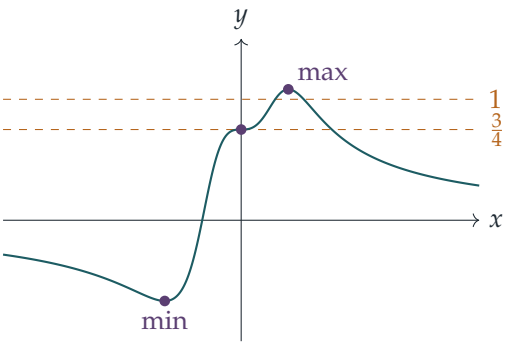
- g is decreasing for $x < \beta$, increasing on (β, α) and decreasing for $x > \alpha$;
- $x = \beta$ is a minimum and $x = \alpha$ a maximum;
- at $x = 0$, $g' > 0$ on both sides, so $x = 0$ is a stationary point of inflection, not a minimum.

Step 4: the global bounds. Even with the correct local maxima and minima, a bound "for all x" needs more: the function could exceed a local maximum somewhere else, for example as $x \to \pm\infty$. Here $g(x) \to 0$ as $x \to \pm\infty$ (the denominator has higher degree), and g is continuous everywhere since $x^4 + 4 > 0$. Combining this with the monotonicity in Step 3, g falls from values near 0 down to $g(\beta)$, rises to $g(\alpha)$, then falls back towards 0. Hence the correct statement is

$$\frac{3}{2\beta} \leq g(x) \leq \frac{3}{2\alpha} \quad \text{for all } x, \qquad \text{i.e. roughly } -0.668 \leq g(x) \leq 1.083,$$

with both bounds attained.

Verdict. The conclusion $\frac{3}{4} \leq g(x) \leq 1$ is false at both ends: $g(-1) = \frac{1}{5} < \frac{3}{4}$ (indeed $g(x) < 0$ whenever $2x^3 + 3 < 0$), and $g(\alpha) > 1$ because $\alpha < \frac{3}{2}$ (check: $h(\frac{3}{2}) = \frac{81}{16} - 3 > 0$, so $\alpha < \frac{3}{2}$ and $\frac{3}{2\alpha} > 1$).



Part (ii)

Step 1: the indefinite integral. The given answer is wrong: differentiating $-3(1-x)^{-4}$ gives $-12(1-x)^{-5}$, not $(1-x)^{-3}$. The power should go *up* by one, from -3 to -2. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}(1-x)^{-2} = -2(1-x)^{-3}\cdot(-1) = 2(1-x)^{-3},$$

the correct result is

$$\int (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}(1-x)^{-2} + c, \qquad x \neq 1.$$

Ask yourself The correct antiderivative also gives $\frac{1}{2}\cdot\frac{1}{4} - \frac{1}{2}\cdot\frac{1}{4} = 0$ between -1 and 3. So is the answer 0 after all?

Step 2: the definite integral. No. Evaluating $[F(x)]_{-1}^{3}$ is justified only when $F' = f$ on the whole interval and f is continuous there. Here $(1-x)^{-3}$ is undefined at $x = 1$, inside $[-1, 3]$, and unbounded near it, so the integral is improper and must be defined as a limit of the two separate pieces:

$$\int_{-1}^{1-\varepsilon} (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}\left(\varepsilon^{-2} - \tfrac{1}{4}\right) \to +\infty \quad (\varepsilon \to 0^{+}),$$
$$\int_{1+\delta}^{3} (1-x)^{-3}\,\mathrm{d}x = \tfrac{1}{2}\left(\tfrac{1}{4} - \delta^{-2}\right) \to -\infty \quad (\delta \to 0^{+}).$$

Both pieces diverge, so the integral does not exist; "$\infty - \infty$" has no value. This also makes sense geometrically: the integrand is positive for $x < 1$ and negative for $x > 1$, and each side encloses infinite area.

Verdict. Both steps are wrong, and the conclusion is false as it stands: $\int_{-1}^{3}(1-x)^{-3}\,\mathrm{d}x$ is undefined. The "0" has a grain of truth only if we cut out a *symmetric* gap, $\varepsilon = \delta$: then the two pieces cancel exactly and the limit is 0 (the Cauchy principal value). But with $\delta = 2\varepsilon$, say, the sum is $\frac{1}{2}(\varepsilon^{-2} - \frac{1}{4}\varepsilon^{-2}) \to \infty$, so the answer depends on how we approach $x = 1$, which is exactly why the integral isn't defined.

Checks

A numerical derivative of g agrees with the factorised formula at random points; $g'(1) = 0.4 \neq 0$. Bisection gives $\alpha \approx 1.3856$, $\beta \approx -2.2467$, and $g(\alpha) = 1.0825 = 3/(2\alpha)$, $g(\beta) = -0.6676 = 3/(2\beta)$; sampling g on $[-100, 100]$ confirms these are the global maximum and minimum. For (ii), midpoint quadrature on $[-1, 0.99]$ gives about 4999.87, matching $\frac{1}{2}(10^4 - \frac{1}{4})$, and the matching piece on $[1.01, 3]$ gives -4999.87.

Common slips

- Deciding the nature of stationary points by comparing their values.
- Forgetting the behaviour as $x \to \pm\infty$ when claiming a global bound.
- Applying $[F(x)]_a^b$ across a point where the integrand is infinite; always check the integrand is continuous on $[a, b]$ first.

Beyond STEP

Part (ii) is about improper integrals: an integral over an interval containing a singularity is defined as a limit, and it converges only if each one-sided limit exists separately. The symmetric limit, the Cauchy principal value, is a genuinely useful idea in complex analysis and physics, but it's a different object. Part (i) is about the gap between local and global extrema, which analysis makes precise via the extreme value theorem. Abbott's *Understanding Analysis* treats both carefully.

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